12-02-25

CSE-815

Probabilistic Reasoning

Event, sample space

$$P(E) + P(F) = P(E \cup F) + P(EF)$$

$S = \{(H,H), (H,T), (T,H), (T,T)\}$

$E = \{(H,H), (H,T)\}$ $\quad F = \{(H,H), (T,H)\}$

Prob. that either head is at first or second coin flip.

$$P(E \cup F) = P(E) + P(F) - P(EF)$$

$$= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$$

$$P(E \cup F) = \{(H,H), (H,T), (T,H)\}$$

Conditional Probability

$\{1, 2, 3, (4), 5, 6\}$ $\quad \{1, 2, 3, 4, 5, 6\}$

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$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

P(sum of two events is 9) = $\frac{4}{36}$

P(first die shows 4) = $\frac{1}{6}$

P(sum of two events is 9 ∩ first die shows 4) = $\frac{1}{36}$

P(sum of two events is 9 | first die shows 4) = $\frac{P(\text{sum of two events is 9} \cap \text{first die shows 4})}{P(\text{first die shows 4})}$

$$= \frac{\frac{1}{36}}{\frac{1}{6}} = \frac{1}{36} \times 6 = \frac{1}{6}$$

Bayesian classification:

Bayes theorem: $P(H|X) = \frac{P(X|H) \cdot P(H)}{P(X)}$ ⇝ $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

posteriori = likelihood × prior / evidence

$$P(C_i|X) = \frac{P(X|C_i) \cdot P(C_i)}{P(X)}$$

$P(X)$ — constant for all C_i

$$P(C_i|X) = P(X|C_i) \cdot P(C_i)$$

to test which class will be maximized

$$P(X|C_i) = \prod_{k=1}^{n} P(x_k|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times \cdots \times P(x_n|C_i)$$

Naive Bayes Classifier: Training Dataset

class:

C1: buys computer = 'yes'

C2: " = 'no'

Data to be classified:

- $X = (x_1$ age <= 30,
- x_2 income = medium,
- x_3 student = yes,
- x_4 credit_rating = Fair)

$$P(C_1) = \frac{9}{14}, \quad P(C_2) = \frac{5}{14}$$

$$P(age <= 30 | C_1) = \frac{2}{9} \rightsquigarrow P(age <= 30 \cap C_1) = \frac{P(age <= 30 \cap C_1)}{P(C_1)}$$

$$P(age <= 30 | C_2) = \frac{3}{5}$$

$$P(income = medium | C_1) = \frac{4}{9}$$

$$P(\quad " \quad | C_2) = \frac{2}{5}$$

$$P(student = yes | C_1) = \frac{6}{9}$$

$$P(\quad " \quad | C_2) = \frac{1}{5}$$

$$P(credit_rating = fair | C_1) = \frac{6}{9}$$

$$P(\quad " \quad | C_2) = \frac{2}{5}$$

$$P(X|C_1) = \frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{6}{9}$$

$$P(X|C_2) =$$

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$P(X|C_i|C_i) \propto P(X|C_1) \times P(X|C_i)$

Since, $P(X|C_1) \times P(C_1) > P(X|C_2) \times P(C_2)$,

so, X belongs to class (buys computer = "yes")

Naive Bayes classification

13-02-25

$$P(C_i|X) = P(X|C_i) \cdot P(C_i) = P(x_1|C_i) \times \dots P(x_n|C_i) \cdot P(C_i)$$

x_1 = Income

x_2 = Student

C_1 — Yes

C_2 — No

$P(C_1) = \frac{3}{6} = \frac{1}{2}$ $\quad P(C_2) = \frac{1}{2}$

$P(x_1 = \text{High} | C_1) = \frac{0}{3} = 0$

$P(x_1 = \text{High} | C_2) = \frac{2}{3}$

$P(x_2 = \text{Yes} | C_1) = \frac{2}{3}$

$P(x_2 = \text{Yes} | C_2) = \frac{0}{3} = 0$

Since we're getting 0, we require correction

Laplacian Correction:

Variation (class) of each attribute is added. (K added to that attribute class)

Corrected likelihood for Income (K=3 since Income can be high, medium or low).

$$P(x_1 = \text{High} \mid C_1) = \frac{0+1}{3+3} = \frac{1}{6}$$

(1: Yes; 3: 3 classes)

$$P(x_1 = \text{High} \mid C_2) = \frac{2+1}{3+3} = \frac{3}{6} = 0.5$$

Corrected likelihood for Student Status (K=2 since Yes, No)

$$P(x_2 = \text{Yes} \mid C_1) = \frac{2+1}{3+2} =$$

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Dataset

Income not categorical like previous problems. Its in form of continuous

ID	Age	Income	student	Buy computer
1	Youth	300	No	No
2	Youth	400	No	No
3	Middle	100	No	Yes
4	Senior	300	Yes	Yes
5	Senior	200	Yes	Yes
6	Youth	300	Yes	No

$P(\text{Age} = \text{Senior} \mid \text{Yes}) = \frac{2+1}{3+3} = \frac{3}{6} = \frac{1}{2}$

$P(\text{Age} = \text{Senior} \mid \text{No}) = \frac{0+1}{3+3} = \frac{1}{6}$

$P(\text{Yes}) = \frac{1}{2}$, $P(\text{No}) = \frac{1}{2}$

$P(\text{Student} = \text{Yes} \mid \text{Yes}) = \frac{2}{3}$

$P(\text{Student} = \text{~~No~~ Yes} \mid \text{No}) = \frac{1}{3}$

x is input which is Income.

We assume it follows Gaussian (Normal) distribution.

If we use normal dist^n no need for connection. Sep?

$$\mu_{yes} = \frac{300+200+100}{3} = 200$$

$$\sigma_{yes} = \frac{\sqrt{(300-200)^2 + (200-200)^2 + (100-200)^2}}{3} = 81.\ldots$$

$$\sigma_{yes}^2 = \frac{(300-200)^2 + (100-200)^2}{3} = 6666.67$$

$$P(\text{Income} = 400 \mid \text{Yes}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left(\frac{x-\mu}{2\sigma^2}\right)}$$

Similarly, $$P(\text{Income} = 400 \mid \text{No}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\left(\frac{x-\mu}{2\sigma^2}\right)}$$

Calculate Posterior Prob.:

$$P(\text{Yes} \mid \text{Age} = \text{Senior}, \text{Income} = 400, \text{Student} = \text{Yes})$$

$$= P(\text{Yes}) \cdot P(\text{Age} = \text{Senior} \mid \text{Yes}) \cdot P(\text{Income} = 400 \mid \text{Yes}) \cdot P(\text{Student} = \text{Yes} \mid \text{Yes})$$

$$P(\text{No} \mid \text{Age} = \text{Senior},$$

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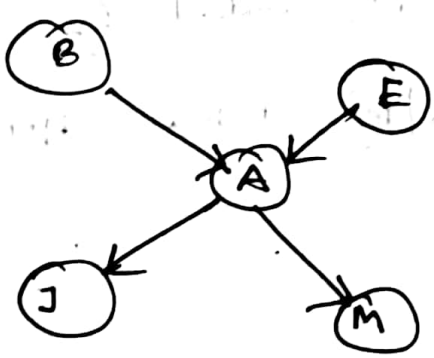
18-02-25

Bayesian Network

- DAG
- random variables & any probabilistic relationship established

Joint Probability Distribution:

$$P(x_1, x_2, x_3) = P(x_1) \cdot P(x_2 | x_1) \cdot P(x_3 | x_1, x_2)$$



$$P(B, E, A, M, J) = P(B) \cdot P(E) \cdot P(A | B, E) \cdot P(J | A) \cdot P(M | A)$$

Conditional Probability Table:

(i) $P(A = Yes \mid B = Yes, E = Yes) = \frac{2}{2} = 1$

(ii) $P(A = Yes \mid B = Yes, E = No) = \frac{2}{2} = 1$

(iii) $P(A = Yes \mid B = No, E = Yes) = \frac{2}{3}$

(iv) $P(A = Yes \mid B = No, E = No) = \frac{1}{3}$

B (Burglary)	E (Earthquake)	A = Yes	A = No
Yes	Yes	1	0
Yes	No	1	0
No	Yes	2/3	1/3
No	No	1/3	2/3

CPT for John calls:

$$P(J = Yes \mid A = Yes) = \frac{5}{7}$$

$$P(J = Yes \mid A = No) = \frac{0}{3} = 0$$

CPT for Mary calls:

$$P(M \mid A)$$

$$P(M = Yes \mid A = Yes) = \frac{5}{7}$$

$$P(M = Yes \mid A = No) = \frac{0}{3} = 0$$

A	J = Yes	J = No

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Now to calculate joint prob:

$$P(B=\text{Yes}, E=\text{No}, A=\text{Yes}, J=\text{Yes}, M=\text{Yes})$$

$$= P(B=\text{Yes}) \cdot P(E=\text{No}) \cdot P(A=\text{Yes} \mid B=\text{Yes}, E=\text{No}) \cdot P(J=\text{Yes} \mid A=\text{Yes}) \cdot P(M=\text{Yes} \mid A=\text{Yes})$$

$$= \frac{2}{10} \cdot \frac{5}{10} \cdot 1 \cdot \frac{6}{7} \cdot \frac{5}{7} = \frac{6}{49}$$

---

① $P(A=\text{Yes}, B=\text{No}, E=\text{No}, J=\text{Yes}, M=\text{No})$

$$= P(B=\text{No}) \cdot P(E=\text{No}) \cdot P(A=\text{Yes} \mid B=\text{No}, E=\text{No}) \cdot P(J=\text{Yes} \mid A=\text{Yes}) \cdot P(M=\text{No} \mid A=\text{Yes})$$

$$= \frac{2}{10} \cdot \frac{5}{10} \cdot \frac{1}{3} \cdot \frac{6}{7} \cdot \frac{2}{7} = \frac{6}{245}$$

② $P(A=\text{No}, B=\text{Yes}, E=\text{Yes}, J=\text{No}, M=\text{No})$

$$= P(B=\text{Yes}) \cdot P(E=\text{Yes}) \cdot P(A=\text{No} \mid B=\text{Yes}, E=\text{Yes}) \cdot P(J=\text{No} \mid A=\text{No}) \cdot P(M=\text{No} \mid A=\text{No})$$

$= \frac{4}{10} \cdot \frac{5}{10} \cdot 0 .$

$= 0$

(iii) $P(A=Yes, B=No, E=Yes, J=Yes, M=No)$

$= P(B=No) \cdot P(E=Yes) \cdot P(A=Yes \mid B=No, E=Yes) \cdot P(J=Yes \mid A=Yes) \cdot P(M=No \mid A=Yes)$

$= 0.04$

06-03-25

Hidden Markov Model

- Probabilistic model
- Time series data
- Hidden state + observation
  - actual states (directly observe করতে পারছো না)

✓ ▷ Evaluation Problem : sequence of observations $O = o_1 o_2 \ldots o_T$. Calculate $P(O|\lambda)$ where HMM λ.

▷ Decoding Problem

▷ Learning Problem

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Evaluation Problem

Using Forward Algorithm

Steps:

1. Initialization: $\alpha_1(i) = \pi_i b_i(o_1)$
2. Recursively compute $\alpha_t(j) = \sum_{i=1}^{N} \alpha_{t-1}(i)\, a_{ij}\, b_j(o_t)$
3. Compute total probability $P(O|\lambda) = \sum_{i=1}^{N} \alpha_T(i)$

Hidden states = {Rainy, Sunny}

Observations = {Walk, Shop, Clean}

Transition Probability Matrix required

where $a_{ij} = P(s_t = j \mid s_{t-1} = i)$

s_t hidden state at time step t

N no. of hidden states.

$$\sum_{j=1}^{N} a_{ij} = 1 \quad \forall i$$

$$A = \begin{array}{c} \\ i=\text{Sunny} \\ i=\text{Rainy} \end{array} \begin{array}{c} \begin{array}{cc} j=\text{Sunny} & j=\text{Rainy} \end{array} \\ \begin{bmatrix} P(\text{Sunny} \mid \text{Sunny}) = \frac{1}{5} & P(\text{Rainy} \mid \text{Sunny}) = \frac{4}{5} \\ P(\text{Sunny} \mid \text{Rainy}) = \frac{3}{4} & P(\text{Rainy} \mid \text{Rainy}) = \frac{1}{4} \end{bmatrix} \end{array}$$

last Rainy ? effect ?

since there is no 4th day.

$$A = \begin{bmatrix} 0.20 & 0.80 \\ 0.75 & 0.25 \end{bmatrix}.$$

Emission Probability matrix

আমি একটা observation state এ থাকার given একটা hidden state আছি

b_{ij} – prob. of observing o_j given hidden state s_i

$i \rightarrow$ hidden state

$j \rightarrow$ observation

$$\sum_{j=1}^{M} b_{ij} = 1$$

$$B = \begin{bmatrix} P(\text{Walk}|\text{Sunny}) = \frac{2}{5} & P(\text{Shop}|\text{Sunny}) = \frac{2}{5} & P(\text{Clean}|\text{Sunny}) = \frac{1}{5} \\ P(\text{Walk}|\text{Rainy}) = \frac{1}{5} & P(\text{Shop}|\text{Rainy}) = \frac{1}{5} & P(\text{Clean}|\text{Rainy}) = \frac{3}{5} \end{bmatrix}$$

$$B = \begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

Initial probabilities

$$\pi = \{ P(\text{Sunny}) = 0.5,\ P(\text{Rainy}) = 0.5 \}$$

Solution:

Initialization

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$$\alpha_1(i) = \pi_i \cdot b_i(o_1)$$

[forward probability for state i at time 1] [initial prob. of state i] [emission probability observing o_1 in state i]

$$\alpha_1(\text{sunny}) = \pi_{\text{sunny}}\, b_{\text{sunny}}(o_1)$$

$$\alpha_1(\text{sunny}) = \pi(\text{sunny}) \cdot P(\text{walk} \mid \text{sunny}) = 0.5 \times 0.4 = 0.2$$

$$\alpha_1(\text{Rainy}) = \pi(\text{Rainy}) \cdot P(\text{walk} \mid \text{Rainy}) = 0.5 \times 0.2 = 0.1$$

Time step ≡ Observation

Recursion

$$\alpha_t(j) = \sum_{i=1}^{N} \alpha_{t-1}(i) \cdot a_{ij} \cdot b_j(o_t)$$

$\alpha_t(j)$: forward prob. for state j at time t

$\alpha_{t-1}(i)$: forward prob. for state i at time t-1

a_{ij}: transition prob. from state i to state j

$b_j(o_t)$: emission prob. of observing o_t in state j

At time t = 2, [second observation - shop] we calculate:

$$\alpha_2(\text{sunny}) = [\alpha_1(\text{sunny}) \cdot P(\text{sunny} \mid \text{sunny}) + \alpha_1(\text{Rainy}) \cdot P(\text{sunny} \mid \text{Rainy})] \cdot P(\text{shop} \mid \text{sunny})$$

$$= (0.2 \times 0.25 + 0.1 \times 0.75) \times 0.4 = 0.046$$

$$\alpha_2(\text{Rainy}) = [\alpha_1(\text{Sunny}) \cdot P(\text{Rainy}|\text{Sunny}) + \alpha_1(\text{Rainy}) \cdot P(\text{Rainy}|\text{Rainy})]$$

$P(\text{Shop}|\text{Rainy})$ — 2nd observation

$$= (0.2 \times 0.80 + 0.1 \times 0.25) \times 0.4$$

$$= 0.074$$

$$\alpha_3(\text{Sunny}) = [\alpha_2(\text{Sunny}) \cdot P(\text{Sunny}|\text{Sunny}) + \alpha_2(\text{Rainy}) \cdot P(\text{Sunny}|\text{Rainy})] \cdot P(\text{clean}|\text{Sunny})$$

3rd observation

$$= (0.046 \times 0.20 + 0.074 \times 0.75) \times 0.4$$

$$= 0.02588$$

$$\alpha_3(\text{Rainy}) = [\alpha_2(\text{Sunny}) \cdot P(\text{Rainy}|\text{Sunny}) + \alpha_2(\text{Rainy}) \cdot P(\text{Rainy}|\text{Rainy})]$$

$P(\text{clean}|\text{Rainy})$

Termination:

$P(O|\lambda)$ – total prob. of observation sequence O

$\alpha_T(i)$ – forward prob for state i at final time step

T

$$P(O|\lambda) = \alpha_3(\text{Sunny}) + \alpha_3(\text{Rainy})$$

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Prob of observing sequence Walk, Shop, clean given HMM

11-03-25

Neural Network

- Best Normalization
- Dropout

ANN

Activation functions:

- Sigmoid function: $\phi(x) = \frac{1}{1+\exp(-x)} = \frac{1}{1+e^{-x}}$
- Step function: $\phi(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \end{cases}$
- Gaussian function: $\phi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{x-\mu}{\sigma}\right)^2}$

Single Layer architecture: Input সরাসরি output এ যাবে

Multi-Layer " : hidden layer থাকবে

* যতটি class ততটি output থাকবে.

Backpropagation

$$I_j = \sum_i w_{ij} O_i + \theta_j$$

$$O_j = \frac{1}{1 + e^{-I_j}}$$

$$Err_j = O_j (1 - O_j)(T_j - O_j)$$

$$Err_j = O_j (1 - O_j) \sum_k Err_k w_{jk}$$

$$\Delta w_{ij} = (l) Err_j O_i$$

$$w_{ij} = w_{ij} + \Delta w_{ij}$$

$$\Delta \theta_j = (l) Err_j$$

$$\theta_j = \theta_j + \Delta \theta_j$$

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l = learning rate

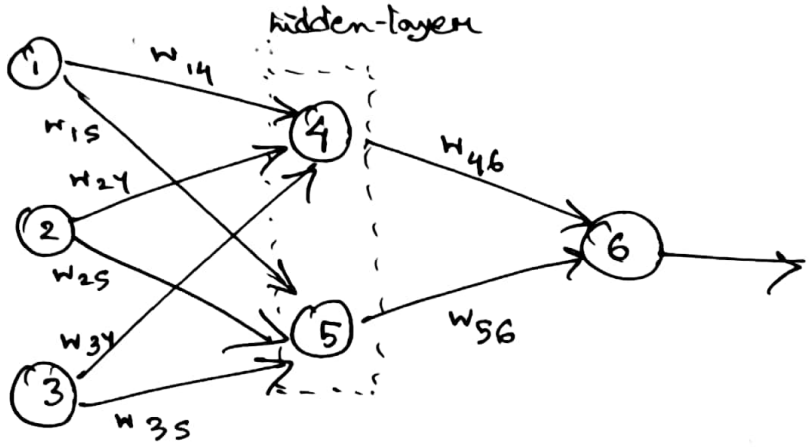


Fig: multi-layer feed-forward neural network

Unit j	Net Input I_j	Output O_j
4	$0.192 \times 1 + 0.4 \times 0 + (-0.508) \times 1 - 0.408$	$1/(1+e^{0.324}) = 0.326$
5	$0.194 \times 1 + 0.1 \times 0 + (-0.306) \times 1 + 0.194$	$1/(1+e^{-0.082}) = 0.520$
6	$0.326 \times (-0.261) + 0.520 \times (-0.138) + 0.218$	$1/(1+e^{-0.051}) = 0.5152$

Epoch = Iteration

$0.5152 \rightarrow 1$

$> 0.5 \rightarrow 1$

$< 0.5 \rightarrow 0$

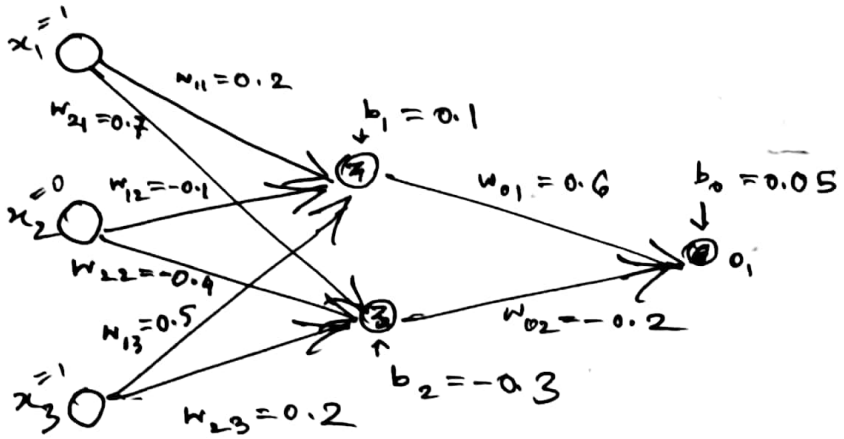
Training data	class
101	1
110	0

Batch Normalization + Dropout (Next class)

12-03-25

$w_{11} = 0.2$ $w_{12} = -0.1$ $w_{13} = 0.5$

$w_{21} = 0.7$ $w_{22} = -0.4$ $w_{23} = 0.2$



Unit j	Next Input I_j	Output O_j
z_1	$1 \times 0.2 + 0 \times (-0.1) + 1 \times 0.5 + 0.1$	$1/(1+e^{-0.8}) = 0.689$
z_2	$1 \times 0.7 + 0 + 0.2 + (-0.3)$	$1/(1+e^{-0.6}) = 0.645$
O_1	$0.689 \times 0.6 + 0.645 (-0.2) + 0.05$	$1/(1+e^{-0.343}) = 0.582$

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Batch Normalization.

Sometimes in hidden layer, variance may be quite high. So to avoid problems in learning, we do batch normalization before activation on hidden layers.

Weighted sum ⟶ Batch Normalization → Activation

$$\mu_B = \frac{1}{m}\sum_{i=1}^{m} z_i$$

$$\sigma_B^2 = \frac{1}{m}\sum_{i=1}^{m}(z_i - \mu_B)^2$$

$$\hat{z_i} = \frac{z_i - \mu B}{\sqrt{\sigma_B^2 + \epsilon}}$$

$$z_i^{BN} = \gamma \hat{z_i} + \beta$$

μ_B – mean of batch

σ_B^2 – variance "

$\hat{z_i}$ – normalized value

ϵ – small no. added for numerical stability

γ – scaling factor

β – shift factor

We use $\gamma = 1$, $\beta = 0$
$\epsilon = 10^{-5}$

$h_1 = 0.690$ ⟵ $z_1 = 0.8$

$h_2 = 0.645$ ⟵ $z_2 = 0.6$

$z_1 = 0.8$, $\mu_B = \frac{1}{2}\cdot(0.8+0.6) = 0.7$

$z_2 = 0.6$

$$\sigma_B^2 = \frac{1}{2}\left\{(0.8-0.7)^2 + (0.6-0.7)^2\right\}$$

$$= 0.01$$

$$\hat{z_1} = \frac{0.8-0.7}{\sqrt{0.01+10^{-5}}}$$

$$= 0.9995$$

$$\hat{z_1} = \frac{0.8-0.7}{\sqrt{(0.01)^2+10^{-5}}} = 9.534$$

$$\hat{z_2} = \frac{0.6-0.7}{\sqrt{0.01+10^{-5}}}$$

$$= -0.9995$$

$$\hat{z_2} = \frac{0.6-0.7}{\sqrt{(0.01)^2+10^{-5}}} = -9.534$$

$$z_1^{BN} = 1 \times 9.534 + 0 = 9.534$$

$$z_2^{BN} = 1 \times (-9.534) + 0 = -9.534$$

$$h_1 = \frac{1}{1+e^{-9.534}} = 0.99992$$

$$h_2 = \frac{1}{1+e^{9.534}} = 7.23\times10$$

$$z_1^{BN} = 0.9995$$

$$z_2^{BN} = -0.9995$$

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~~$h_0 =$~~

$$h_1 = \frac{1}{1+e^{-0.9995}} = 0.730$$

$$h_2 = \frac{1}{1+e^{0.9995}} = 0.269$$

For hidden to output layers

$$\mu_B = 0.4995$$

$$\sigma_B^2 = \frac{1}{2}\left((0.4995-0.730)^2 + (0.4995-0.269)^2\right)$$

$$= 0.053$$

$$z_1 = \frac{0.730 - 0.4995}{\sqrt{0.053 + 10^{-5}}} = 1.0011$$

$$z_2 = \frac{0.269 - 0.4995}{\sqrt{0.053 + 10^{-5}}} = -1.0011$$

$z_1^{BN} = 1.0011$, $z_2^{BN} = -1.0011$

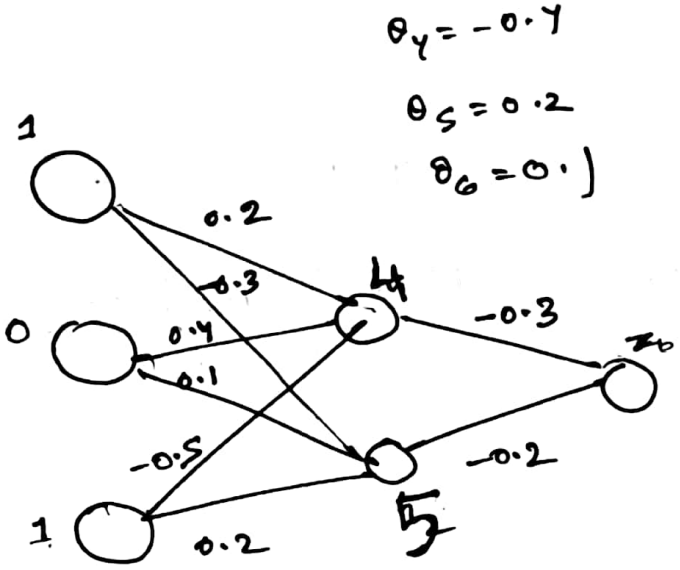
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Dropout

Prevents overfitting.

Process:

- During each training step, dropout (20-50%) to 0.



Combine Batch Normalization with Dropout

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1. Batch Normalization

Pre-activation values

$z_1 = 0.2 \times 1 + (-0.5) \times 1 - 0.4 = -0.7$

$z_2 = -0.3 + 0.2 + 0.2 = 0.1$

$$\mu_B = \frac{1}{2}(-0.7 + 0.1) = -0.3$$

$$\sigma_B^2 = \frac{1}{2}\left((-0.7+0.3)^2 + (0.1+0.3)^2\right)$$

$$= 0.16$$

$$\hat{z}_1 = \frac{-0.7 + 0.3}{\sqrt{\sigma_B^2 + \epsilon}} = \frac{-0.4}{\sqrt{0.16 + 10^{-5}}} = -0.999968$$

$\sigma_B^2 = 0.16$, $\epsilon = 10^{-5}$

$$\hat{z}_2 = \frac{0.1 + 0.3}{\sqrt{\sigma_B^2 + \epsilon}} = \frac{0.4}{\sqrt{0.16 + 10^{-5}}} = 0.999968$$

$$z_1^{BN} = \gamma \hat{z}_1 + \beta = 1 \times (-0.999968) + 0$$

$$= -0.999968$$

$$z_2^{BN} = 0.999968$$

$$h_1^{BN} = \sigma(z_1^{BN}) = \frac{1}{1 + e^{-(-0.999968)}}$$

$$= 0.2689$$

$$h_2^{BN} = \sigma(z_2^{BN}) = \frac{1}{1 + e^{0.999968}}$$

$$= 0.73105$$

$$z_0 = -0.3 \times h_1^{BN} + (-0.2) \times h_2^{BN} + \theta_0$$

$$\therefore = -0.3 \times 0.2689 + (-0.2) \times 0.73105 + 0.1 = -0.12688$$

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$$\hat{y} = \sigma(z_0) = \frac{1}{1+e^{-(-0.1268)}} = 0.468$$

4. Dropout

Preactivation values $z_1 = -0.7$

$z_2 = 0.1$

Sigmoid activations $h_1 = \sigma(z_1) = \frac{1}{1+e^{0.7}} = 0.3318$

$h_2 = \sigma(z_2) = \frac{1}{1+e^{-0.1}} = 0.5249$

Assuming 5 is dropped, $h_2 = 0$, so h_1 stays 0.3318

$$z_0 = h_1 \times (-0.3) + h_2 \times (-0.2) + 0.1$$

$$= 0.3318 \times (-0.3) + 0.1$$

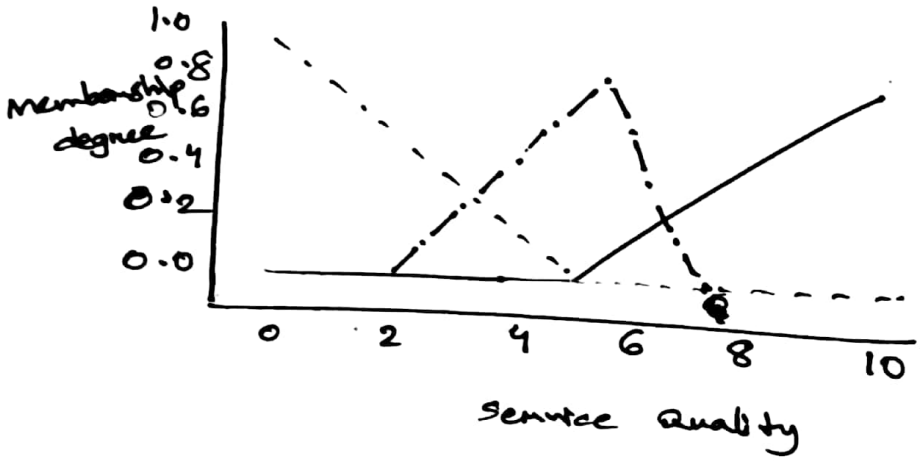
$$= 4.6 \times 10^{-4}$$

$$\hat{y} = \frac{1}{1+e^{-4.6\times10^{-4}}} = 0.5001$$

Fuzzy Logic

Takagi:- Sugeno-Kang (TSK) Approach to Tip calculation

1st step: Input & output Fuzzification



AND এলাকে minimum
OR এলাকে maximum

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#Problem:

① Service Quality = 3
Food quality = 4

② Service Quality = 65
Food quality = 1

$$\mu(x) = \begin{cases} \frac{x-a}{b-a} & a \le x \le b \\ \frac{c-x}{c-b} & b \le x \le c \\ 0 & \text{otherwise.} \end{cases}$$

Service Quality (3):

Poor: a = 0, b = 0, c = 5

Avg: a = 2, b = 5, c = 8

Excellent: a = 5, b = 10, c = 10

For x = 3;

- Poor:

$$\mu_{poor}(3) = \frac{3-0}{5-0} = \frac{3}{5}$$

- Avg:

~~$\mu_{avg}(3) = 0$ [outside range]~~

~~$\mu_{avg}(3) = 0$ [out~~

$$\mu_{avg}(3) = \frac{3-2}{5-2} = \frac{1}{3}$$

- Excellent:

$$\mu_{excellent}(3) = 0 \quad [\text{outside range } [5, 10]]$$

Food quality (4):

- Bad : a=0, b=0, c=5
- Good : a=2, b=5, c=8
- Great : a=5, b=10, c=10

for x = 4,

- Bad:

$$\mu_{Bad}(4) = \frac{4-0}{5-0} = \frac{4}{5}$$

~~$\mu_{Bad}(8) = 0$ [outside range]~~

- Good:

~~$\mu_{Good}(8) = \frac{8-8}{8-5} = 0$~~

$$\mu_{Good}(4) = \frac{8-4}{8-5} = \frac{4}{3}$$

- Great:

$$\mu_{Great}(4) = 0 \quad [\text{outside range}]$$

Rule Activation:

Rule 1: Low Tip is activated with

$$\mu_{Rule1} = \max(\mu_{service\,Poor}, \mu_{Food\,Bad})$$

$$= \max\left(\frac{3}{5}, \frac{4}{5}\right) = \frac{4}{5}$$

Rule 2: Medium Tip is activated with

$$\mu_{Rule2} = \min(\mu_{service\,Avg}, \mu_{Food\,Good})$$

$$= \min\left(\frac{1}{3}, \frac{4}{3}\right)$$

$$= \frac{1}{3}$$

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Rule 3: High Tip activated with

$$\mu_{Rule\,3} = \max\left(\mu_{service\,Excellent}, \mu_{Food\,Great}\right)$$

$$= \max(0,0) = 0$$

Apply Rule Outputs:

Rule 1: $\mu_{Rule\,1} = \frac{4}{5}$, $output\ function_1 = 5$

Weighted $Output_1 = 4$

Rule 2:

$\mu_{Rule\,2} = \frac{1}{3}$, $output\ function_2 = 10 + \frac{1}{2}\cdot 3 + \frac{1}{2}\cdot 4^2$

$= 12 + \frac{3}{2} = \frac{27}{2}$

Weighted $Output_2 = \frac{1}{3}\cdot\frac{27}{2} = \frac{9}{2}$

Rule 3:

$\mu_{Rule\,3} = 0$, $output\ function_3 = 15$

Weighted $Output_3 = 0$.

Aggregate Rule outputs:

Total Weighted $Output = 4 + \frac{9}{2} = \frac{17}{2}$

Total weight $= \frac{4}{5} + \frac{1}{3} + 0$

$= \frac{17}{15}$

$\therefore$ Cuisp Output = $\frac{\text{Total Weighted output}}{\text{Total weight}} = \frac{\frac{17}{2}}{\frac{17}{15}}$

$= \frac{1/2}{1/6} = 6/5/4$

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chapter - 8

23-04-25

Decision Tree Induction

Historical database থেকে analyze করে decision tree generate করা।

Example 8.4

$X = (age = youth, income = medium, student = yes, credit_rating = fair)$

Attribute Selection Measures:

Technique :-

1. Information Gain :

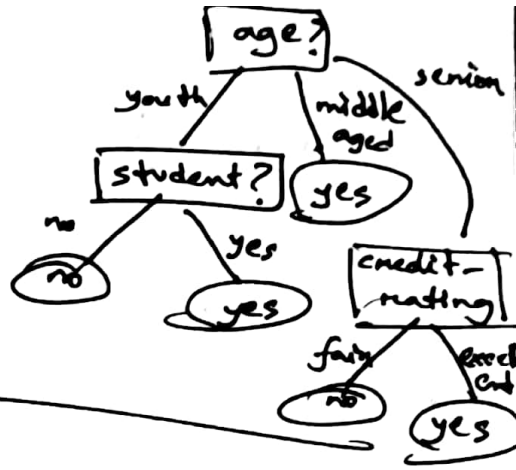
$$Info(D) = -\sum_{i=1}^{m} p_i \log_2(p_i)$$

p_i — nonzero probability that an arbitrary tuple in D belongs to class C_i. Estimated by $|C_{i,D}|/|D|$.

$Info(D)$ - entropy

$$Info_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j)$$

$$Gain(A) = Info(D) - Info_A(D)$$



$$Info(D) = -\frac{9}{14}\log_2\left(\frac{9}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 0.940 \text{ bits}$$

(9: 9 yes buying comps; 5: no; 14: total data)

$$Info_{age}(D) = \frac{5}{14} \times \left(-\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}\right) +$$

(5: 5 youths; 2: 2 yes in youth; 3: 3 no in youth; 5: মোট youth)

$$\frac{4}{14} \times \left(-\frac{4}{4}\log_2\frac{4}{4}\right) +$$

(4: 4 middle aged → 4 yes buying comps)

$$\frac{5}{14} \times \left(-\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}\right)$$

$$= 0.694 \text{ bits}$$

$$Gain(age) = Info(D) - Info_{age}(D)$$

$$= 0.246 \text{ bits}.$$

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$$\text{Info}_{\text{income}}(D) = \frac{4}{14} \times \left(-\frac{2}{4}\log_2\frac{2}{4} - \frac{2}{4}\log_2\frac{2}{4}\right) + \quad \text{(high)}$$

$$\frac{6}{14} \times \left(-\frac{4}{6}\log_2\frac{4}{6} - \frac{2}{6}\log_2\frac{2}{6}\right) + \quad \text{(medium)}$$

$$\frac{4}{14} \times \left(-\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4}\right) \quad \text{(low)}$$

$$= \frac{4}{14} \times \left(\overset{\times\frac{2}{4}}{1} + \overset{\times\frac{2}{4}}{1}\right) + \frac{6}{14}\left(\overset{\times\frac{4}{6}}{0.585} + \overset{\times\frac{2}{6}}{1.585}\right)$$

$$+ \frac{4}{14}\left(-\frac{3}{4}\times(-0.415) - \frac{1}{4}\times(-2)\right)$$

$$= 0.285$$

~~$= 0.142$~~ ~~$= 0.142$~~ $+ 0.393 + 0.231$

$= 0.909$

~~$= 0.766$~~ ⟶

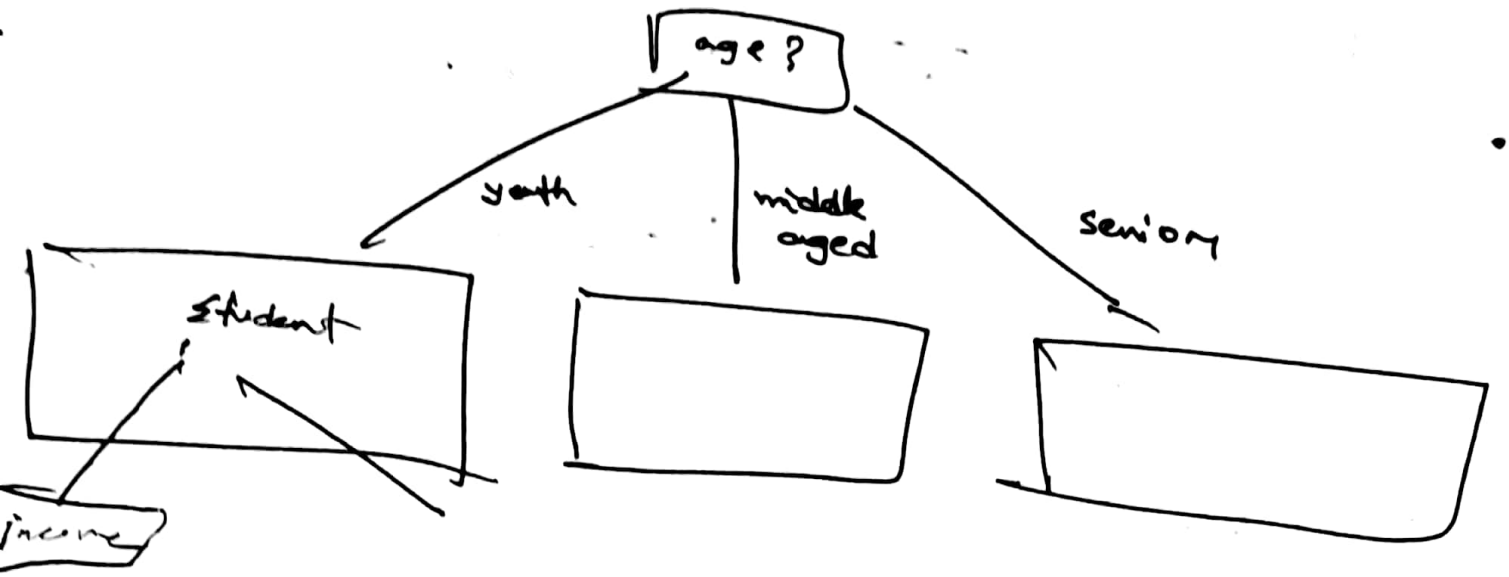
$\therefore$ Gain (income) $= 0.940 - 0.766$ bits

~~$= 0.174$ bits~~ $= 0.030$ bits.

Similarly, Gain (student) $= 0.151$ bits

Gain (credit_rating) $= 0.048$ bits

Since age has highest info gain, it is selected as splitting attribute.



on youth,

$$\text{Info}(D) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$= -\frac{3}{5}\times(-0.737) - \frac{2}{5}\times(-1.322)$$

$$= 0.971 \text{ bits}$$

$$\text{Info}_{\text{income}}(D) = \frac{2}{5}\times\left(0 - \frac{2}{2}\log_2\frac{2}{2}\right) +$$

$$\frac{2}{5}\times\left(-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}\right) +$$

$$\frac{1}{5}\times\left(-\frac{1}{1}\log_2\frac{1}{1} - 0\right)$$

$$= \frac{2}{5}\times 0 + \frac{2}{5}\left(\frac{1}{2} + \frac{1}{2}\right) +$$

$$\frac{1}{5}\times 0 = \frac{2}{5} = 0.4 \text{ bits}$$

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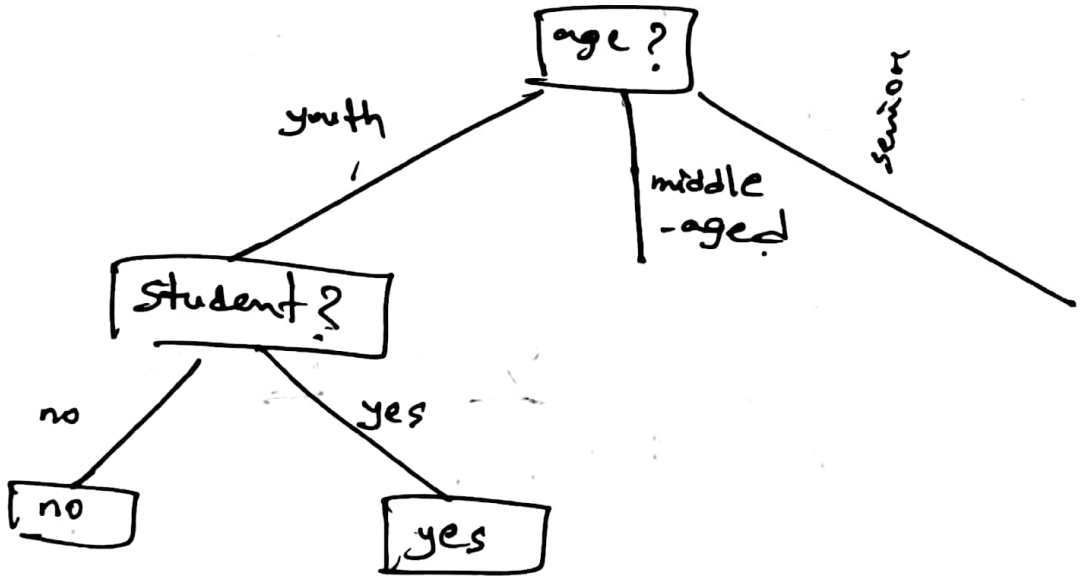
$\therefore$ Gain (Income) $\geq$ 0.971 $-$ 0.4 = 0.571 bits

$$\text{Info}_{\text{student}}(D) = \frac{2}{5} \times \left(-\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\right) + \frac{3}{5} \times \left(-0 - \frac{3}{3}\log_2\frac{3}{3}\right)$$

$$= 0$$

$\therefore$ Gain (student) $= 0.971 - 0 = 0.971$ bits

Since student has the highest info gain in the subtree below age, so it will be the splitting attribute for subtree.



For middle aged,

$$\text{Info}(D) = -\frac{4}{4}\log_2\frac{4}{4} - 0$$

$$= -1 \times 0 - 0 = 0$$

Since $\text{Info}(D) = 0$ for subtree of middle aged,

For senior,

$$\text{Info}(D) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.9709$$

$$\text{Info}_{\text{student}}(D) = \frac{3}{5} \times \left(-\frac{3}{3}\log_2\frac{3}{3} - \frac{1}{3}\log_2\frac{1}{3}\right) + \frac{2}{5} \times \left(-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}\right)$$

$$=$$

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Eqn - 9.5

$$\text{Split Info}_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right)$$

$$\text{Gain Ratio}(A) = \frac{\text{Gain}(A)}{\text{Split Info}_A(D)}$$

Gain Ratio of Age, student, Income, Credit_rating.

Given, Gain Ratio (income) = 0.019

Gain (age) = 0.246,

Gain (student) = 0.151,

Gain (credit_rating) = 0.048.

$$\text{Split Info}_{age}(D) = \frac{-5}{14} \cdot \log_2\left(\frac{5}{14}\right) - \frac{4}{14}\log_2\left(\frac{4}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right)$$

$$= 0.530 + 0.516 + 0.530$$

$$= 1.576$$

$$\therefore \text{Gain Ratio (age)} = \frac{0.246}{1.576}$$

$$= 0.156$$

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$$\text{SplitInfo}_{\text{student}}(D) = -\frac{7}{14}\log_2\left(\frac{7}{14}\right) - \frac{7}{14}\log_2\left(\frac{7}{14}\right)$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

$$\therefore \text{Gain Ratio (student)} = \frac{0.151}{1}$$

$$= 0.151$$

$$\text{SplitInfo}_{\text{credit_rating}}(D) = -\frac{6}{14}\log_2\left(\frac{6}{14}\right) - \frac{8}{14}\log_2\left(\frac{8}{14}\right)$$

$\frac{6}{14}\log_2\left(\frac{6}{14}\right) = .5238$, $\frac{8}{14}\log_2\left(\frac{8}{14}\right) = .461$

$$= 0.985$$

$$\therefore \text{Gain Ratio (credit_rating)} = \frac{.048}{.985}$$

$$= 0.0487$$

Since GainRatio (age) is the highest, so it is the splitting attribute.

29-04-25

Model Evaluation and Selection

Confusion matrix – used to evaluate performance of classification model.

Fig-8.13

Accuracy, Recognition rate $= \frac{TP + TN}{P+N}$

Error rate, misclassification rate $= \frac{FP + FN}{P+N}$

Sensitivity, True Positive Rate, Recall $= \frac{TP}{P}$

Specificity, True Negative Rate $= \frac{TN}{N}$

Precision $= \frac{TP}{TP + FP}$

F, F_1, F-score,
Harmonic mean of Precision & Recall $= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

Fig-8.14

Predicted class

Actual class	yes	no	
yes	TP	FN	P
no	FP	TN	N
Total	P'	N'	P+N



classes	buy-computer=y	buy-comp=no	Total	Recognition
buy-computer=y	6954	46	7000	99.34
buy-computer=n	412	2588	3000	86.27
	7366	2634	10,000	95.42

∴ Recognition rate = Accuracy $= \frac{6954 + 2588}{7000 + 3000} = 95.42\%$

∴ Error rate $= \frac{412 + 46}{7000 + 3000} = 4.58\%$

∴ Sensitivity $= \frac{6954}{7000} = 99.34\%$

∴ Specificity $= \frac{2588}{3000} = 86.26\%$

∴ Precision $= \frac{6954}{6954 + 412} = 94.40\%$.

* Determine precision & recall for each class, And F_1 score.

		Predicted A	Predicted B	Predicted C
Actual	A	30	10	10
	B	5	40	5
	C	3	5	42

Precision for class A, $P_A = \frac{30}{30 + 5 + 3}$

$= 0.789$

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$$P_B = \frac{40}{40+10+5} = 72.72\%$$

$$P_C = \frac{42}{42+10+5} = 73.68\%$$

$$\text{Accuracy} = \frac{30+40+42}{150}$$

Recall, $R_A = \frac{30}{30+10+10} = \frac{30}{50}$

$$R_B = \frac{40}{50}$$

$$R_C = \frac{42}{3+5+42} = \frac{42}{50}$$

F_1 score for A $= \frac{2 \times 0.7894 \times \frac{30}{50}}{.7894 + \frac{30}{50}} = 0.6818$

" " B $= \frac{2 \times .7272 \times \frac{40}{50}}{.7272 + \frac{40}{50}} = 0.7619$

" " C $= \frac{2 \times .7368 \times \frac{42}{50}}{.7368 + \frac{42}{50}} = 0.7850$

class	A	A	A	A	A	B	B	B	B	B
Data ID	~~2~~	~~3~~	~~4~~	~~5~~	~~6~~	~~7~~	7	8	9	10
	1	2	3	4	5	6				

i. Holdout

ii. Random subsampling

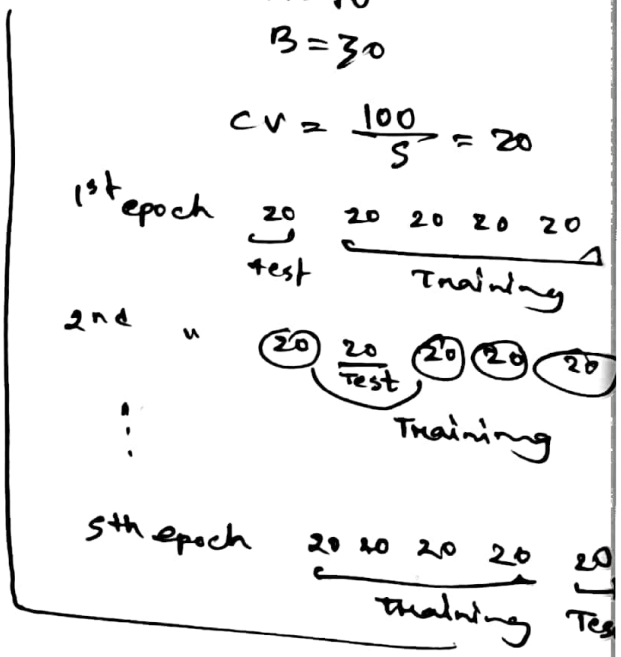
iii. ~~Stra~~ Cross validation

iv. Stratified Cross Validation

14 - A
6 - B

20 20 20 20 20

A = 70
B = 30



08-05-25

Cluster Analysis

K-means clustering → partitioning / hierarchical method

$K \geq 2$

Point #	x	y	
1	3	4	
2	2	1	→ c1
3	5	2	
4	6	2	
5	1	4	
6	7	2	→ c2
7	2	7	
8	8	5	

First, arbitrarily choose cluster centre. We use 2 since $K=2$.

	c_1	c_2
1	$\sqrt{10}$	$\sqrt{20}$
2	$\sqrt{10}$	$\sqrt{26}$
3	$\sqrt{17}$	2
4	$\sqrt{10}$	
5	$\sqrt{26}$	
6	6	
7	$\sqrt{52}$	
8		

	c_1	c_2
1	$\sqrt{10}$	$\sqrt{20}$
2		$\sqrt{26}$
3	$\sqrt{10}$	2
4	$\sqrt{17}$	1
5	$\sqrt{10}$	$\sqrt{40}$
6	$\sqrt{26}$	
7	6	$\sqrt{50}$
8	$\sqrt{52}$	$\sqrt{10}$

Euclidean dist^n

point #	x	y	c_1	c_2
1	3	4	3.16	4.47
2	2	1	0	5.1
3	5	2	3.16	2
4	6	2	4.12	1
5	1	4	3.16	6.32
6	7	2	5.09	0
7	2	7	6	7.07
8	8	5	7.24	3.16

$c_1 - 1, 2, 5, 7$

$c_2 - 3, 4, 6, 8$

New cluster center

$$= \left(\frac{3.16 + 0 + 3.16 + 6}{4}, \frac{2 + 1 + 0 + 3.16}{4} \right)$$

$$= \left(\frac{3+2+1+2}{4}, \frac{4+1+4+7}{4} \right)$$

$= 3.08$

$c1_{new} = 2, 4$

$c2_{new} = 6.5, 2.75$

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(3, 4) (6.5, 2.75)

Point	x	y	c_1 new	c_2 new
1	3	4	1	3.7165
2	2	1	3	~~4.8283~~
3	5	2	3.605	1.6770
4	6	2	4.4721	0.9013
5	1	4	1	~~2.1565~~ 5.64
6	7	2	5.3851	0.9013
7	2	7	3	6.1897
8	8	5	~~6.0827~~ ~~2.7077~~ 6.0827	~~7.3125~~ 2.7041

c_1 – 1, 2, 5, 7

c_2 – 3, 4, 6, 8

Same clusters obtained after 1st iteration, so we stop

If successive cluster center are same stop iteration.